Day 28 - [8.2.2 | 8.2.3] Complex Numbers

Thursday, March 27, 2025 9:12 AM

Objective:
- Students will be able to…
  - investigate properties of Polynomial equations
  - determine what makes a graph bounce, wiggle, or cross the x-axis
  - determine the degree of a polynomial

Mathematical Practice(s): reason abstractly and quantitatively, attend to precision, look for and make use of structure

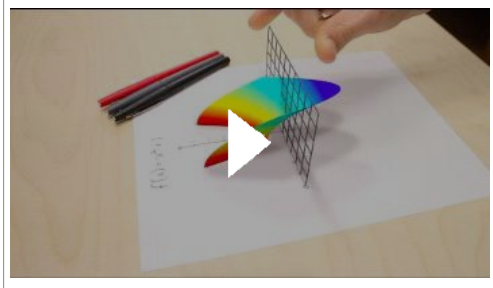
Classwork:
- Warm-Up
- Video & Debrief Questions
- Graphing Imaginary Numbers
  - Model How to Label Axes & Complete 8-97 A
  - 8-99
- Fundamental Thm of Alg
  - 8-100
- What is i?
  - 8-65
- Quadratics with Imaginary Roots
  - 8-66 & 8-78
  - Model 8-86 A

Homework:
- Day 29 HW

Announcement:
- Day 29: Trigonometry KC #2
  - ODD - Wednesday, April 15th
  - EVEN - Thursday, April 16th

Video

Imaginary Numbers Are Real [Part 1: Introduction]



Summary

- **Notation for Polynomials**

  The **general equation** of a second-degree (quadratic) polynomial is often written in the form $f(x) = ax^2 + bx + c$, and the general equation of a third-degree (cubic) polynomial is often written in the form $f(x) = ax^3 + bx^2 + cx + d$.

  For a polynomial with an undetermined degree n, it is unknown how many letters will be needed for the coefficients.
  Instead of using a, b, c, d, e, etc., mathematicians use only the letter a, and they used subscripts, as shown below.

  $$\boldsymbol{f(x) = (a_n)x^n + (a_{n-1})x^{n-1} + \ldots + (a_1)x^1 + a_0}$$

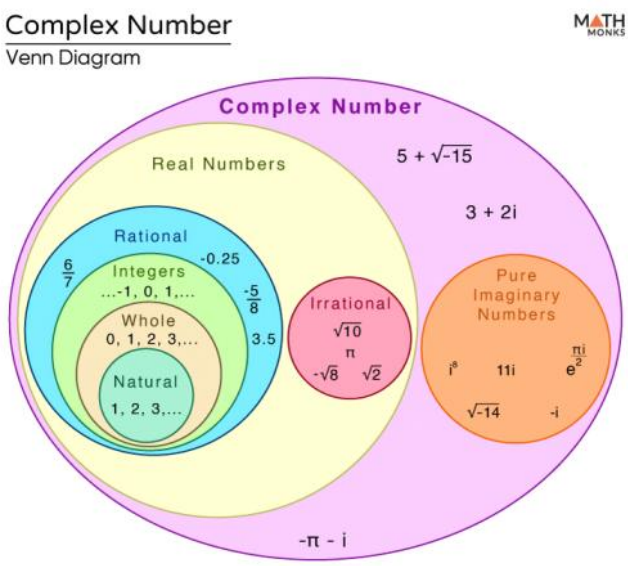
  This general polynomial has degree n and coefficients $a_n, a_{n-1}, \ldots a_1, a_0$.

- **Complex Numbers → a + bi**
  - A Complex Number can be expressed in the form $z = a + bi$ where a and b are real numbers and i is the imaginary unit.
  - This form is known as the Cartesian Form of a Complex Number where "cartesian" essentially relates to the form being able to help us plot or graph on a 2D or 3D plane. Similar to "cartesian coordinates" or (x,y) coordinates.

  - Important:
    - a, b are both real values
    - $i = \sqrt{-1}$
    - a is the real part of the complex number
    - b is the imaginary part of the complex number
    - $a + bi$ is called the Cartesian Form of a Complex Number
    - If real part, a, is equal to zero, then bi is considered a pure imaginary

For example, for $7x^4 - 5x^3 + 3x^2 + 7x + 8$, the degree is 4.
In this specific case:
a_n is a_4 *and* $a_4 = 7$
a_{n-1} *is* $a_3 = -5$
a_{n-2} *is* $a_2 = 3$
$a_1 = 7$
$a_0 = 8$.

Complex Number System



- **Imaginary number i**
  - $i = \sqrt{-1}$
  - $i^2 = -1$
  - $\frac{1}{i} = -i$
  - Operations
    - **Adding | Subtracting** - Combine Like Terms (treat i like a variable)
    - **Multiplying** - Operate as if i is a variable & simplify all i^2 terms by replacing with -1
    - **Dividing** - See above for special case dividing by i. Otherwise, we treat i similar to that of a root, we want to "rationalize" it out of the denominator.